

Worked HOTA Example — 4 Frames, 3 GT Tracks, 4 Predicted Tracks

Below numeric walk-through below uses a **single threshold**

$$\alpha = 0.5$$

while the real HOTA code repeats the thresholded part for all

$$\alpha \in \{0.05, 0.10, \dots, 0.95\}.$$

1. Track presence across 4 frames

GT tracks

GT track	Frame 1	Frame 2	Frame 3	Frame 4	Total detections
G_1	✓	✓	✓	✓	4
G_2	✓	✓	–	✓	3
G_3	✓	✓	✓	✓	4

So:

$$N_{G_1}^G = 4, \quad N_{G_2}^G = 3, \quad N_{G_3}^G = 4.$$

Predicted tracks

Pred track	Frame 1	Frame 2	Frame 3	Frame 4	Total detections
P_1	✓	✓	✓	✓	4
P_2	✓	✓	✓	✓	4
P_3	✓	–	✓	✓	3
P_4	–	✓	–	✓	2

So:

$$N_{P_1}^P = 4, \quad N_{P_2}^P = 4, \quad N_{P_3}^P = 3, \quad N_{P_4}^P = 2.$$

2. Per-frame similarity matrices S^t

Per-frame Raw and Normalized Similarity Matrices

For each frame, the raw similarity matrix S^t is converted into the normalized similarity matrix R^t using

$$R_{ij}^t = \frac{S_{ij}^t}{\sum_j S_{ij}^t + \sum_i S_{ij}^t - S_{ij}^t}.$$

The row labels correspond to GT tracks and the column labels correspond to predicted tracks.

Frame 1

$$S^1 = \begin{array}{c|ccc} & P_1 & P_2 & P_3 \\ \hline G_1 & 0.90 & 0.60 & 0.00 \\ G_2 & 0.20 & 0.55 & 0.00 \\ G_3 & 0.00 & 0.30 & 0.85 \end{array} \xrightarrow{\text{row/column normalization}} R^1 = \begin{array}{c|ccc} & P_1 & P_2 & P_3 \\ \hline G_1 & 0.52 & 0.25 & 0.00 \\ G_2 & 0.12 & 0.33 & 0.00 \\ G_3 & 0.00 & 0.13 & 0.73 \end{array}$$

For example,

$$R_{G_1, P_1}^1 = \frac{0.90}{1.50 + 1.10 - 0.90} = \frac{0.90}{1.70} = 0.5294$$

whereas

$$R_{G_1, P_2}^1 = \frac{0.60}{1.50 + 1.45 - 0.60} = \frac{0.60}{2.35} = 0.2553.$$

Although $S_{G_1, P_2}^1 = 0.60$ is reasonably high, the normalized score is reduced because P_2 has significant similarity with multiple GT objects.

In contrast,

$$R_{G_3, P_3}^1 = \frac{0.85}{1.15 + 0.85 - 0.85} = \frac{0.85}{1.15} = 0.7391,$$

showing that (G_3, P_3) is a strong and relatively unambiguous correspondence.

Frame 2

$$S^2 = \begin{array}{c|ccc} & P_1 & P_2 & P_4 \\ \hline G_1 & 0.88 & 0.52 & 0.00 \\ G_2 & 0.10 & 0.58 & 0.45 \\ G_3 & 0.00 & 0.35 & 0.50 \end{array} \xrightarrow{\text{row/column normalization}} R^2 = \begin{array}{c|ccc} & P_1 & P_2 & P_4 \\ \hline G_1 & 0.58 & 0.22 & 0.00 \\ G_2 & 0.04 & 0.29 & 0.27 \\ G_3 & 0.00 & 0.17 & 0.38 \end{array}$$

Here, P_2 overlaps with all three GT objects, so its normalized similarities are reduced. The $G_1 \leftrightarrow P_1$ correspondence remains comparatively strong and clean.

Frame 3

$$S^3 = \begin{array}{c|ccc} & P_1 & P_2 & P_3 \\ \hline G_1 & 0.40 & 0.72 & 0.00 \\ G_3 & 0.00 & 0.40 & 0.83 \end{array} \xrightarrow{\text{row/column normalization}} R^3 = \begin{array}{c|ccc} & P_1 & P_2 & P_3 \\ \hline G_1 & 0.35 & 0.47 & 0.00 \\ G_3 & 0.00 & 0.20 & 0.67 \end{array}$$

Although G_1 has similarity with both P_1 and P_2 , normalization favors P_2 . For G_3 , the P_3 correspondence is substantially cleaner than P_2 .

Frame 4

$$S^4 = \begin{array}{c|cccc} & P_1 & P_2 & P_3 & P_4 \\ \hline G_1 & 0.86 & 0.50 & 0.00 & 0.00 \\ G_2 & 0.00 & 0.48 & 0.00 & 0.62 \\ G_3 & 0.00 & 0.20 & 0.80 & 0.42 \end{array} \xrightarrow{\text{row/column normalization}} R^4 = \begin{array}{c|cccc} & P_1 & P_2 & P_3 & P_4 \\ \hline G_1 & 0.63 & 0.24 & 0.00 & 0.00 \\ G_2 & 0.00 & 0.26 & 0.00 & 0.40 \\ G_3 & 0.00 & 0.08 & 0.56 & 0.20 \end{array}$$

The normalization again suppresses ambiguous predictions such as P_2 , while relatively isolated correspondences such as $G_1 \leftrightarrow P_1$ and $G_3 \leftrightarrow P_3$ retain higher normalized scores.

Overall, the transformation

$$S^t \xrightarrow{\text{row/column normalization}} R^t$$

rewards **strong, unambiguous GT-prediction correspondences** while penalizing similarities that compete with several alternatives.

3. Accumulate global identity evidence C_{gp}

$$C_{gp} = \sum_{t,i,j:ID_G(i)=g, ID_P(j)=p} R^t_{ij}$$

Non-zero values:

$$C_{G_1,P_1} = 2.10, \quad C_{G_1,P_2} = 1.19$$

$$C_{G_2,P_1} = 0.17, \quad C_{G_2,P_2} = 0.89, \quad C_{G_2,P_4} = 0.68$$

$$C_{G_3,P_2} = 0.59, \quad C_{G_3,P_3} = 1.97, \quad C_{G_3,P_4} = 0.59$$

C_{gp}	P_1	P_2	P_3	P_4
G_1	2.10	1.19	0	0
G_2	0.17	0.89	0	0.68
G_3	0	0.59	1.97	0.59

4. Global identity alignment A_{gp}

$$A_{gp} = \frac{C_{gp}}{N^G_g + N^P_p - C_{gp}}$$

Examples:

$$A_{G_1,P_1} = \frac{2.10}{4 + 4 - 2.10} = 0.35$$

$$A_{G_1,P_2} = \frac{1.19}{4 + 4 - 1.19} = 0.17$$

$$A_{G_3,P_3} = \frac{1.97}{4 + 3 - 1.97} = 0.39$$

Global alignment matrix:

A_{gp}	P_1	P_2	P_3	P_4
G_1	0.35	0.17	0	0
G_2	0.02	0.14	0	0.15
G_3	0	0.08	0.39	0.10

5. Frame-level matching-score matrices W^t

$$W_{ij}^t = A_{ID_G(i), ID_P(j)} \cdot S_{ij}^t$$

Frame 1

$$W^1 = \begin{bmatrix} 0.32 & 0.10 & 0.00 \\ 0.00 & 0.08 & 0.00 \\ 0.00 & 0.02 & 0.33 \end{bmatrix}$$

Hungarian:

$$\Pi_1 = \{(G_1, P_1), (G_2, P_2), (G_3, P_3)\}$$

Since all selected pairs have $S \geq 0.5$:

$$\Pi_1^{0.5} = \Pi_1$$

Frame 2

$$W^2 = \begin{bmatrix} 0.31 & 0.09 & 0.00 \\ 0.00 & 0.08 & 0.07 \\ 0.00 & 0.02 & 0.05 \end{bmatrix}$$

(Columns correspond to P_1, P_2, P_4 .)

Hungarian:

$$\Pi_2 = \{(G_1, P_1), (G_2, P_2), (G_3, P_4)\}$$

Again all selected pairs satisfy $S \geq 0.5$, so:

$$\Pi_2^{0.5} = \Pi_2$$

Frame 3

$$W^3 = \begin{bmatrix} 0.14 & 0.13 & 0.00 \\ 0.00 & 0.03 & 0.33 \end{bmatrix}$$

(Columns correspond to P_1, P_2, P_3 .)

Hungarian:

$$\Pi_3 = \{(G_1, P_1), (G_3, P_3)\}$$

Thresholding with $\alpha = 0.5$:

- (G_1, P_1) : raw similarity $0.40 < 0.5 \rightarrow$ reject
- (G_3, P_3) : raw similarity $0.83 \geq 0.5 \rightarrow$ accept

So:

$$\Pi_3^{0.5} = \{(G_3, P_3)\}$$

Even though $S_{G_1, P_2}^3 = 0.72$ is larger than $S_{G_1, P_1}^3 = 0.40$, the global alignment makes (G_1, P_1) score slightly higher in W^3 .

Frame 4

$$W^4 = \begin{bmatrix} 0.31 & 0.09 & 0.00 & 0.00 \\ 0.00 & 0.07 & 0.00 & 0.10 \\ 0.00 & 0.02 & 0.31 & 0.05 \end{bmatrix}$$

Hungarian:

$$\Pi_4 = \{(G_1, P_1), (G_2, P_4), (G_3, P_3)\}$$

All chosen pairs satisfy $S \geq 0.5$, hence:

$$\Pi_4^{0.5} = \Pi_4$$

6. Summary of matches

Frame	Hungarian Π_t	Accepted $\Pi_t^{0.5}$
1	$(G_1, P_1), (G_2, P_2), (G_3, P_3)$	same
2	$(G_1, P_1), (G_2, P_2), (G_3, P_4)$	same
3	$(G_1, P_1), (G_3, P_3)$	(G_3, P_3) only
4	$(G_1, P_1), (G_2, P_4), (G_3, P_3)$	same

7. Detection counts at $\alpha = 0.5$

Per frame:

- Frame 1: $TP = 3, FN = 0, FP = 0$
- Frame 2: $TP = 3, FN = 0, FP = 0$
- Frame 3: $TP = 1, FN = 1, FP = 2$
- Frame 4: $TP = 3, FN = 0, FP = 1$

Overall:

$$TP_{0.5} = 10, \quad FN_{0.5} = 1, \quad FP_{0.5} = 3$$

8. Identity-pair match counts $M_{gp}^{0.5}$

Accepted identity-pair counts:

$$\begin{aligned} M_{G_1, P_1}^{0.5} &= 3 \\ M_{G_2, P_2}^{0.5} &= 2, \quad M_{G_2, P_4}^{0.5} = 1 \\ M_{G_3, P_3}^{0.5} &= 3, \quad M_{G_3, P_4}^{0.5} = 1 \end{aligned}$$

Distributed match-count matrix:

$M_{gp}^{0.5}$	P_1	P_2	P_3	P_4
G_1	3	0	0	0
G_2	0	2	0	1
G_3	0	0	3	1

9. Association IoU per identity pair

$$AssIoU_{gp}^{\alpha} = \frac{M_{gp}^{\alpha}}{N_g^G + N_p^P - M_{gp}^{\alpha}}$$

At $\alpha = 0.5$:

$$AssIoU_{G_1, P_1}^{0.5} = \frac{3}{4 + 4 - 3} = 0.60$$

$$AssIoU_{G_2, P_2}^{0.5} = \frac{2}{3 + 4 - 2} = 0.40$$

$$AssIoU_{G_2, P_4}^{0.5} = \frac{1}{3 + 2 - 1} = 0.25$$

$$AssIoU_{G_3, P_3}^{0.5} = \frac{3}{4 + 3 - 3} = 0.75$$

$$AssIoU_{G_3, P_4}^{0.5} = \frac{1}{4 + 2 - 1} = 0.20$$

Distributed match-count matrix:

$AssIoU_{gp}^{0.5}$	P_1	P_2	P_3	P_4
G_1	0.6	0	0	0
G_2	0	0.4	0	0.25
G_3	0	0	0.75	0.20

10. Association Accuracy $AssA_{0.5}$

$$AssA_{0.5} = \frac{\sum_{g,p} M_{gp}^{0.5} AssIoU_{gp}^{0.5}}{TP_{0.5}}$$

$$AssA_{0.5} = \frac{3(0.60) + 2(0.40) + 1(0.25) + 3(0.75) + 1(0.20)}{10}$$

$$AssA_{0.5} = \frac{1.80 + 0.80 + 0.25 + 2.25 + 0.20}{10} = \frac{5.30}{10} = 0.53$$

11. Localization Accuracy $LocA_{0.5}$

Accepted-similarity sum:

- Frame 1: $0.90 + 0.55 + 0.85 = 2.30$
- Frame 2: $0.88 + 0.58 + 0.50 = 1.96$
- Frame 3: 0.83
- Frame 4: $0.86 + 0.62 + 0.80 = 2.28$

Therefore:

$$L_{0.5} = 2.30 + 1.96 + 0.83 + 2.28 = 7.37$$

$$LocA_{0.5} = \frac{L_{0.5}}{TP_{0.5}} = \frac{7.37}{10} = 0.737$$

12. Detection metrics

$$DetRe_{0.5} = \frac{10}{10 + 1} = 0.9091$$

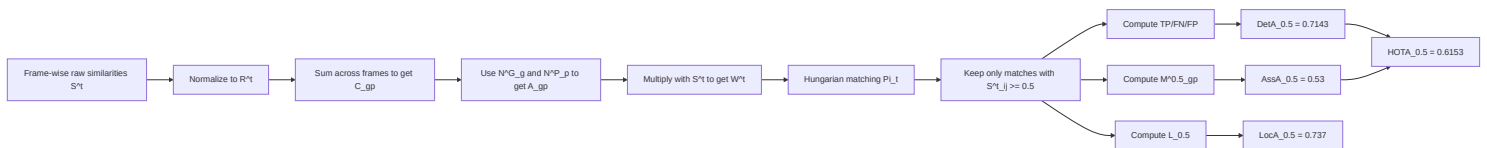
$$DetPr_{0.5} = \frac{10}{10 + 3} = 0.7692$$

$$DetA_{0.5} = \frac{10}{10 + 1 + 3} = \frac{10}{14} = 0.7143$$

13. Final HOTA at $\alpha = 0.5$

$$HOTA_{0.5} = \sqrt{DetA_{0.5} \cdot AssA_{0.5}} = \sqrt{0.7143 \cdot 0.53} = 0.6153$$

14. Compact “calculation flow” summary



15. Key takeaways

1. Normalization matters

Ambiguous row/column overlap reduces normalized contribution.

2. Global alignment matters

Matching is based on $W^t = A \odot S^t$, not just raw similarity.

3. matches_count is distributed

Here $M_{gp}^{0.5}$ is not a perfect diagonal.

4. Association and detection both affect HOTA

Imperfect track consistency reduces $AssA$, which reduces final HOTA.